

STA 640 — Causal Inference

Chapter 3.2: Observational Studies

- Stratification and Matching

Fan Li

Department of Statistical Science
Duke University

Balancing covariates: small number of covariates

- ▶ When the number of covariates is small, the adjustment can be achieved by exact matching or stratification
- ▶ Exact matching: for each treated subject, get a control with exact same value of the covariate
- ▶ Exact matching ensures distributions of covariates in treatment and control groups are exactly the same, thus eliminate bias due to difference in X .
- ▶ Exact matching is usually infeasible, even with low dimensional covariates

Stratification

- ▶ Suppose we have a single covariate X with k levels (e.g. age in tens) and X makes assignment ignorable. Want to estimate the ATE τ .
- ▶ We have: $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y|X_i = k, Z = 1)Pr(X = k)$
- ▶ n_k : number of people in cell $X_i = k$; $\bar{Y}_{k,z}$: the sample average of Y among people in the cell $X_i = k$ and $Z_i = z$
- ▶ Estimate $\mathbb{E}[Y(1)]$ by a consistent estimator $\sum_k \bar{Y}_{k,1} \frac{n_k}{n}$.
Therefore, τ can be estimated by:

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n} \quad (1)$$

- ▶ Note: Equation (1) is **not** generally applicable for a **non-linear** contrast of the potential outcomes.

Stratification

- ▶ What if X is continuous?
- ▶ Stratification (subclassification): split X into k classes. Then for class k , define $n_k, \hat{Y}_{k,z}$ as before. An estimator of ATE τ is:

$$\hat{\tau}^k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}$$

- ▶ $\hat{\tau}^k$ is generally **biased** for τ (Lunceford and Davidian, 2004)
- ▶ Denote $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}^k) - \tau}{\mathbb{E}(\hat{\tau}^1) - \tau}$. Cochran (1968) showed that **$R^k \approx 90\%$ for $k \geq 5$** for a large class of underlying models
- ▶ Therefore, generally stratification of over 5 blocks can remove 90% of the bias
- ▶ Not as popular as matching or weighting

Propensity score stratification

- ▶ With multiple covariates, there are too many strata
- ▶ Recall Cochran's result of 5 subclasses of a single covariate removing 90% bias: **using propensity score as the single summary score to stratify** (Rosenbaum and Rubin, 1984)
- ▶ Divide the subjects in to K strata by the corresponding quantiles of the estimated propensity score.
- ▶ Estimate ATE within each stratum of e_k , ($k = 1, \dots, K$) and then average by the stratum size:

$$\hat{\tau}^{\text{ATE}} = \sum_{k=1}^K \{(\bar{Y}_{k,1} - \bar{Y}_{k,0})\} \frac{N_{k,1} + N_{k,0}}{N},$$

with $N_{k,1}, N_{k,0}$ being the numbers of units in class k under trt and control, respectively.

- ▶ My experience: 5 blocks is usually not enough, consider higher number such as 10.

Propensity score stratification

- ▶ Variance via bootstrap or estimator:

$$Var(\hat{\tau}) = \sum_{k=1}^K \{(\bar{Y}_{k,1}) + (\bar{Y}_{k,0})\} \left(\frac{N_{k,1} + N_{k,0}}{N} \right)^2$$

- ▶ Stratification is a coarsened version of matching and weighting; good for cases with extreme outliers (smoothing): less sensitive
- ▶ Stratification estimator is **asymptotically biased** (Lunceford and Davidian, 2004), less efficient than weighting or matching
- ▶ Stratification: first PS method Rosenbaum and Rubin (1984), but has become less popular
- ▶ Combine with regression: first estimate causal effects using regression within each stratum and then average the within-stratum estimates

Matching

- ▶ Regression estimators impute the missing potential outcomes using the estimated regression function
- ▶ Matching estimators also impute the missing potential outcomes, using the outcomes of nearest (in terms of a certain **distance measure**) neighbours of the opposite treatment group
- ▶ Matching is similar to nonparametric kernel regression, **essentially an imputation method**
- ▶ They have often (but not exclusively) been applied in settings where
 - ▶ target estimand is ATT
 - ▶ a large reservoir of potential controls, allowing matching each treated unit to one or more distinct controls (matching without replacement)
- ▶ More general settings: both treated and control units are (potentially) matched and matching is done with replacement

1 – M Nearest-Neighbor (NN) Matching

- ▶ let $\mathcal{M}_i$ be the set of the indices of the M closest matches of unit i in terms of the distance measure based on the norm $\|\cdot\|$.
- ▶ Impute missing potential outcomes

$$\hat{Y}_i(0) = \begin{cases} \sum_{j \in \mathcal{M}_i} Y_j / M, & Z_i = 1, \\ Y_i, & Z_i = 0. \end{cases} \quad \hat{Y}_i(1) = \begin{cases} Y_i, & Z_i = 1, \\ \sum_{j \in \mathcal{M}_i} Y_j / M, & Z_i = 0. \end{cases}$$

- ▶ Matching estimator: treatment effect within a pair is estimated as the difference in outcomes, and then average the within-pair differences

$$\hat{\tau}_{\text{ATE}}^m = \sum_i (\hat{Y}_i(1) - \hat{Y}_i(0)) / N,$$

- ▶ Pros: ensure covariate balance between groups; generally robust
- ▶ Cons: With fixed number of matches and matching **with replacement**, the NN matching estimators are generally **biased**, the asymptotic bias is of the order $O(N^{-1/p})$, where p is the number of continuous covariates (Abadie and Imbens, 2006)

Nearest-Neighbor (NN) Matching: Variance Estimation

- ▶ With fixed number of matches and matching **with replacement**, bootstrap estimate of s.e. of $1 - M$ NN matching estimators is generally **biased** (Abadie and Imbens, 2008)
- ▶ **Intuition:** bootstrap fails to reproduce the distribution of the number of times each unit is used as a match
- ▶ Abadie and Imbens (2006): Normal approximation to asymptotic variance of NN matching estimators
- ▶ Matching estimators are generally **not efficient**, estimators combining matching and regression adjustment are usually more efficient

Bias-corrected matching: combine matching with regression

- ▶ Combine matching with regression: (i) correct for bias (due to residual imbalance) in matching; (ii) increase efficiency
- ▶ Old idea in Rubin (1973): bias correction via regression on the matched sample
- ▶ Let $\mu_z(x) = \mathbb{E}[Y(z)|X = x]$, and $\hat{\mu}_z(X_i)$ be a regression estimator of $\mu_z(X_i)$, e.g., OLS fit, for $z = 0, 1$.
- ▶ Abadie and Imbens (2011, JBES)
 - ▶ derived estimator for the bias of $\hat{\tau}^m$: $\hat{B} = \sum_i^N \hat{B}_i / N$, with

$$\hat{B}_i = (2Z_i - 1) \frac{1}{M} \sum_{j \in \mathcal{M}_i} [\hat{\mu}_{1-Z_i}(X_i) - \hat{\mu}_{1-Z_i}(X_j)]$$

For a treated unit, the estimated bias is $\hat{B}_i = \sum_{j \in \mathcal{M}_i} [\hat{\mu}_0(X_i) - \hat{\mu}_0(X_j)]$

- ▶ Proposed bias-corrected matching estimator: $\hat{\tau}^{mbc} = \hat{\tau}^m - \hat{B}$

Bias-corrected matching: combine matching with regression

Lin, Ding, Han (2023, Econometrica)

- ▶ $\hat{\tau}^{mbc} = \sum_i^N \hat{\psi}_i / N$, where

$$\hat{\psi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + (2Z_i - 1) \left(1 + \frac{K_i}{M} \right) [Y_i - \hat{\mu}_{Z_i}(X_i)]$$

where K_i is the times that unit i is used as a match

- ▶ Sample variance estimator of $\hat{\tau}^{mbc}$:

$$\hat{V}^{mbc} = \frac{1}{N^2} \sum_i^N (\hat{\psi}_i - \hat{\tau}^{mbc})^2$$

- ▶ $\hat{\tau}^{mbc}$ is double-robust:

$$\hat{\tau}^{mbc} = \hat{\tau}^{reg} + \sum_i^N \left[\left(1 + \frac{K_i}{M} \right) Z_i \hat{R}_i - \left(1 + \frac{K_i}{M} \right) (1 - Z_i) \hat{R}_i \right]$$

because as $M \rightarrow \infty$

$$1 + \frac{K_i}{M} \rightarrow \begin{cases} \frac{1}{e(X_i)}, & Z_i = 1, \\ \frac{1}{1-e(X_i)}, & Z_i = 0. \end{cases}$$

Matching: Lots of Tuning Decisions

- ▶ Many matching methods besides NN matching are available
- ▶ Matching involves many tuning decisions
 - ▶ distance metric: Mahalanobis distance, propensity score, variance distributional discrepancies
 - ▶ high-dimensional covariates: covariate selection, order of importance of covariates
 - ▶ fixed or varying no. matches
 - ▶ for fixed M , number of matches
 - ▶ with or without replacement
 - ▶ caliper or nearest neighbor
 - ▶ how to deal with ties
- ▶ In practice, choice of specific matching method is more of a personal preference

Matching: Tuning

- ▶ Fixed M or varying M ? For varying M :
 - ▶ Matching with caliper: define a caliper (say 0.1) and all units within that caliper are matches
 - ▶ M increases with sample size (e.g. kernel-based matching (Heckman et al. 1998))
- ▶ For fixed M , the choice of M (number of matches per unit) has a bias-variance tradeoff: smaller M , smaller bias but larger variance; larger M , larger bias but smaller variance
- ▶ Also depends on the proportion of treatment versus control: when there is a much larger control group, can use 1-to-many matching

Matching: Tuning

- ▶ Matching with replacement:

Pros: (1) computationally easier, (2) both controls and treated can be matched, but with **high variances**, and (3) not order-dependent;

Cons: some units (especially ones with extreme ps) can be matched many times and thus heavily influence overall estimates, similar to extreme weights in weighting

- ▶ Matching is a vast topic: an excellent review up to 2010 is Stuart (2010), but many new matching methods have been developed since (e.g. Rosenbaum and students)
- ▶ Software in R: `Matching` (Sekhon) (this one is faster and easier to use by my experience); `Matchit` (Ho et al.), and many more
- ▶ My experience and recommendation: **most matching methods lead to similar results; in applications, just pick an easy-to-implement method (say default in popular packages) and proceed**

Matching: Target Population?

- ▶ Even with the same sample, different matching methods may lead to very different matched samples, which may corresponds to very different subpopulation
- ▶ So what is the target population (the one causal effect is defined on)?
Can be ambiguous
- ▶ It is irrelevant to compare different matching methods corresponding to different target populations, when treatment effect is heterogenous
- ▶ Also applying the same matching method to different samples can lead to very different target populations and conclusions. E.g. samples with different proportions of treatment and control
- ▶ Always ask ahead what is your target population?

Distance Metric: Mahalanobis distance

- ▶ Mahalanobis distance (Mahalanobis, 1936).
- ▶ For two random vector $\mathbf{x} = (x_1, x_2, \dots, x_p)'$ and $\mathbf{y} = (y_1, y_2, \dots, y_p)'$ from the the same distribution with the covariance matrix S , the Mahalanobis distance is:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})' S^{-1} (\mathbf{x} - \mathbf{y})}.$$

- ▶ If the covariance matrix is the identity matrix, the Mahalanobis distance reduces to the *Euclidean distance*
- ▶ If the covariance matrix is diagonal, reduces a *normalized Euclidean distance*

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^p (x_i - y_i)^2 / s_i^2},$$

where s_i is the standard deviation of x_i (and y_i)

- ▶ Widely applicable, but **computational intensive for high dimensions**

Dimensional Reduction

- ▶ What if there is a large number of covariates? With just 20 binary covariates, there are 2^{20} or about a million covariate patterns
- ▶ Direct matching or stratification is nearly impossible
- ▶ Need dimensional reduction: propensity score

Matching as a Pre-Processing Step

- ▶ More generally, matching can be used as a pre-processing step for causal inference with observational data
 - ▶ Use a generic matching method to obtain a matched sample (nothing optimal), to ensure good balance
 - ▶ Then use regression (e.g. bias-corrected, or simply use the regression coefficient as causal estimate) or weighting on the matched sample to correct for residual imbalance and improve efficiency
- ▶ Empirical results (e.g. on the famous Lalonde data) show that once a matched sample with reasonable balance is obtained, the point estimates from different additional methods are usually similar
- ▶ In high dimensional cases, combining matching (or stratification or weighting) with regression is the way to go (i.e. augmented or double learning)
- ▶ Matching is the starting point, not the end

References

- Abadie, A., Imbens, G. W. (2006). Large sample properties of matching estimators for average treatment effects. *Econometrica*, 74(1), 235-267.
- Abadie, A., Imbens, G. W. (2008). On the failure of the bootstrap for matching estimators. *Econometrica*, 76(6), 1537-1557.
- Abadie, A., Imbens, G. W. (2011). Bias-corrected matching estimators for average treatment effects. *Journal of Business and Economic Statistics*, 29(1), 1-11.
- Austin, P. C. (2011). An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate behavioral research*, 46(3), 399-424.
- Cochran, W. G. (1968). The effectiveness of adjustment by subclassification in removing bias in observational studies. *Biometrics*, 295-313.
- Heckman, J. J., Ichimura, H., Todd, P. (1998). Matching as an econometric evaluation estimator. *The review of economic studies*, 65(2), 261-294.
- Z Lin, P Ding, F Han. (2023). Estimation based on nearest neighbor matching: from density ratio to average treatment effect. *Econometrica* 91 (6), 2187-2217
- Mahalanobis, P. C. (1936). On the generalized distance in statistics. National Institute of Science of India.
- Rubin, D. B. (1973). The use of matched sampling and regression adjustment to remove bias in observational studies. *Biometrics*, 185-203.
- Rubin, D. B. (1976). Multivariate matching methods that are equal percent bias reducing, I: Some examples. *Biometrics*, 109-120.
- Stuart, E. A. (2010). Matching methods for causal inference: A review and a look forward. *Statistical science*, 25(1), 1.